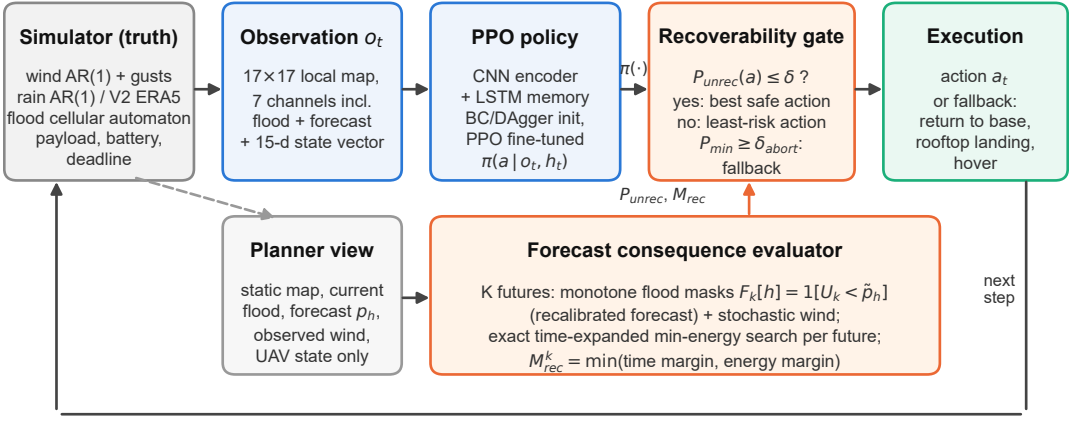

Simulator (truth)
wind AR(1) + gusts
rain AR(1) / V2 ERA5
flood cellular automaton
payload, battery,
deadline
Observation o_t
17×17 local map,
7 channels incl.
flood + forecast
+ 15-d state vector
PPO policy
CNN encoder
+ LSTM memory
BC/DAgger init,
PPO fine-tuned
$\pi(a \mid o_t, h_t)$
$\pi(\cdot)$
Recoverability gate
$P_{unrec}(a) \le \delta$?
yes: best safe action
no: least-risk action
$P_{min} \ge \delta_{abort}$:
fallback
Execution
action a_t
or fallback:
return to base,
rooftop landing,
hover
P_{unrec}, M_{rec}
Planner view
static map, current
flood, forecast p_h,
observed wind,
UAV state only
Forecast consequence evaluator
K futures: monotone flood masks $F_k[h] = 1[U_k < \tilde{p}_h]$
(recalibrated forecast) + stochastic wind;
exact time-expanded min-energy search per future;
M^k_{rec} = min(time margin, energy margin)
next
step